SCIE1500: Analytical Methods for Scientists

SAMPLE FINAL EXAM

The University of Western Australia

Purpose of this document. This sample exam has been provided so that you have a clear understanding of the type, style, and difficulty of questions you will encounter in the final exam. It is based on the Semester 1, 2026 exam and has been extended to include a new Python section (Part II) that reflects the Semester 2, 2026 exam format.

Exam format. The actual exam consists of three parts:

- **Part I — Multiple Choice** (72 marks; 2 marks each): 36 questions on mathematical methods and science applications. Answer on the MCQ answer sheet.
- **Part II — Python as a Mathematical Tool** (14 marks; 2 marks each): 7 multiple-choice questions testing your ability to use Python to perform mathematical tasks at a conceptual level. *You will not need a computer*: questions are designed to be answerable from your understanding of how Python tools work. Answer on the MCQ answer sheet.
- **Part III — Long Answer** (48 marks; 12 marks each): 4 application questions. Show all working in the answer booklet.

Permitted materials. You will *not* be provided with a formula sheet for the final exam, but you are allowed to bring **2 double-sided A4 sheets** of handwritten or printed notes. Include in those notes any formulas you may need but cannot easily recall (e.g. rules of differentiation, integration, Bayes' theorem, binomial formula). A UWA-approved scientific calculator is permitted; programmable calculators, graphing calculators, and mobile phones are **not** permitted. Please check with your lab instructor before the exam day that your calculator model is approved to avoid arriving without one.

This is an examination WITH permitted materials

Provided by the University
1 x 18 Page Answer Booklet
1 x MCQ Answer Sheet (Buff 11556) 125 questions, Answers A-E
Graph Paper - 1 mm

Supplied by the Student
2 x double-sided A4 page of notes (printed or handwritten)

Calculator
UWA approved calculators with stickers are permitted

Preparation advice. To prepare for the final exam, use the sample questions below together with the materials you have worked through during semester: lesson content and practice questions in the SciQuant app; lab exercises and worksheets; and assignment questions and feedback. The Python questions reward students who genuinely engaged with the weekly lab notebooks.

PART I — Multiple Choice Questions (72 marks; 2 marks per question)

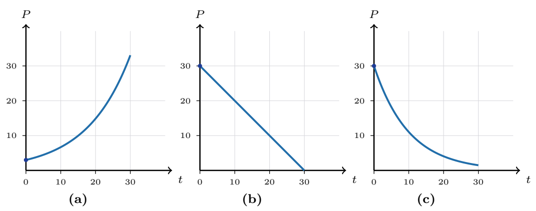
Answer ALL multiple choice questions using the MCQ answer sheet.

1. Which of the following functions has its domain *incorrectly* identified?
   (a) $y = 2 + 5x,\ D = \mathbb{R}$
   (b) $y = \dfrac{\sqrt{3x-6}}{4} - 1,\ D = \mathbb{R}$
   (c) $y = \dfrac{3}{1-e^{-x}},\ D = \{x \in \mathbb{R} : x \neq 0\}$
   (d) $y = e^{-0.2x+1},\ D = \mathbb{R}$
   (e) None of the above.

2. Let $f(x) = -e^{-3x+2}$ and $g(x) = -\frac{1}{3}x^2$. Which of the following is the composite function $(f \circ g)(x)$?
   (a) $-e^{x^2+2}$
   (b) $-e^{-x^2+2}$
   (c) $\frac{1}{3}e^{-3x+2}$
   (d) $e^{-x^2+2} - 2$
   (e) None of the above.

3. Consider the quadratic function $y = -x^2+6x-5$. Which of the following statements about this function is *NOT* true?
   (a) The vertex of the parabola is at $(3, 4)$.
   (b) The y-intercept is at $y = -5$.
   (c) The x-intercepts are at $x = 1$ and $x = 5$.
   (d) The range is $\{y \in \mathbb{R} : y \geq 4\}$.
   (e) None of the above.

4. A bacterial population grows exponentially according to $P(t) = 3\,e^{0.08t}$, where t is measured in hours. Which of the following plots best represents $P(t)$ on the interval $0 \leq t \leq 30$?



   (a) Plot (a).
   (b) Plot (b).
   (c) Plot (c).
   (d) None of the plots shown.
   (e) Cannot be determined from the information given.

5. A population doubles every 2.5 years under continuous exponential growth $P(t) = P_0\,e^{rt}$. What is the continuous per-year growth rate r?
   (a) $r \approx 0.40$ (40.0%)
   (b) $r \approx 0.50$ (50.0%)
   (c) $r \approx 0.277$ (27.7%)
   (d) $r \approx 0.693$ (69.3%)
   (e) None of the above.

6. A population follows logistic growth
$$P(t) = \frac{K}{1 + A\,e^{-\alpha t}}, \qquad K = 20{,}000,\ A = 80,\ \alpha = 0.20.$$
   Approximately how many time periods does it take for the population to reach 50% of the carrying capacity?
   (a) $t \approx 12$
   (b) $t \approx 18$
   (c) $t \approx 22$
   (d) $t \approx 40$
   (e) None of the above.

7. Consider the Schaefer surplus production model $G(S) = g\,S\left(1 - \dfrac{S}{K}\right)$. Which of the following statements is *NOT* true?
   (a) The maximum sustainable yield occurs at $S = K/2$.
   (b) $G(S) \geq 0$ for all $0 \leq S \leq K$.
   (c) The per-capita growth rate $G(S)/S$ declines linearly with S.
   (d) The second derivative $G''(S)$ is positive, so G is concave up.
   (e) All of the above are true.

8. The inverse function of $f(x) = 5 + \frac{1}{3}x$ is
   (a) $f^{-1}(x) = -15 + 3x$
   (b) $f^{-1}(x) = \dfrac{1}{5 + x/3}$
   (c) $f^{-1}(x) = \frac{1}{3}x - 5$
   (d) $f^{-1}(x) = 3x + 15$
   (e) None of the above.

9. Find the equation of the tangent line to the curve $y = e^{x-3} + 2$ at the point where $x = 3$.
   (a) $y = x$
   (b) $y = x - 3$
   (c) $y = e\,x$
   (d) $y = x + 2$
   (e) None of the above.

10. If $y = \frac{1}{2}x^4 + \frac{1}{3}x^3 - 2x + \ln x$, then $\dfrac{dy}{dx}$ equals
   (a) $2x^3 + x^2 - 2 + \dfrac{1}{x}$
   (b) $2x^4 + x^3 - 2x + \ln x$
   (c) $\frac{1}{2}x^3 + \frac{1}{3}x^2 - 2 + \dfrac{1}{x}$
   (d) $2x^3 + x^2 - 2 + x$

(e) None of the above.

11. If $y = x^2 e^{-x}$, then $\dfrac{dy}{dx}$ equals
   (a) $2x\, e^{-x}$
   (b) $-x^2 e^{-x}$
   (c) $x\, e^{-x}(2-x)$
   (d) $e^{-x}(2x+x^2)$
   (e) None of the above.

12. Use the chain rule to differentiate $y = (3x^2+1)^4$. Which of the following is $\dfrac{dy}{dx}$?
   (a) $4(3x^2+1)^3$
   (b) $24x\,(3x^2+1)^3$
   (c) $6x(3x^2+1)^4$
   (d) $12x(3x^2+1)^3$
   (e) None of the above.

13. Consider the function $f(x) = -2x^2 + 12x + 5$. At which value of x does f attain its maximum, and what is the maximum value?
   (a) $x = 2,\ f(2) = 21$
   (b) $x = 3,\ f(3) = 23$
   (c) $x = 6,\ f(6) = 5$
   (d) $x = -3,\ f(-3) = -49$
   (e) None of the above.

14. The indefinite integral $\displaystyle\int (4x^3 - 2x + 5)\, dx$ equals
   (a) $x^4 - x^2 + 5x + C$
   (b) $12x^2 - 2 + C$
   (c) $x^4 - 2x + 5x + C$
   (d) $4x^4 - x^2 + 5x + C$
   (e) None of the above.

15. Suppose $F'(x) = 3x^2 + 2$ and $F(1) = 5$. Then $F(x)$ equals
   (a) $x^3 + 2x + 2$
   (b) $x^3 + 2x$
   (c) $x^3 + 2x + 5$
   (d) $6x + 2$
   (e) None of the above.

16. The area under the curve $y = \ln x$ between $x = 3$ and $x = 15$ is
   (a) $15\ln(15) - 3\ln(3) - 12$
   (b) $15\ln(15) - 3\ln(3)$
   (c) $\ln(15) - \ln(3)$
   (d) $\frac{1}{2}(15^2 - 3^2)$
   (e) None of the above.

17. In a supply-and-demand diagram, the consumer surplus at the market equilibrium (P^*, Q^*) is
   (a) The area between the supply curve and the price line, above the equilibrium price.
   (b) The area under the demand curve from $Q = 0$ to $Q = Q^*$, minus total expenditure P^*Q^*.
   (c) The area between the demand and supply curves to the right of equilibrium.
   (d) Total revenue to producers at equilibrium, P^*Q^*.
   (e) None of the above.

18. Consider a Lotka-Volterra predator-prey model

$$\dot{H} = \alpha H - \beta HP,$$
$$\dot{P} = \lambda HP - \gamma P,$$

with $\alpha = 0.10$, $\beta = 0.004$, $\lambda = 0.0004$, $\gamma = 0.08$. Which of the following statements is *NOT* true?
   (a) $(H, P) = (0, 0)$ is a fixed point.
   (b) The non-trivial equilibrium is $(H^*, P^*) = (200, 25)$.
   (c) The predator conversion efficiency λ/β is 10%.
   (d) The predator conversion efficiency λ/β is 0.04%.
   (e) Trajectories in the (H, P) phase plane are closed orbits.

19. In the Lotka-Volterra system above, if the prey intrinsic growth rate α doubles while all other parameters stay fixed, what happens to the equilibrium *predator* level P^*?
   (a) P^* doubles.
   (b) P^* halves.
   (c) P^* is unchanged.
   (d) P^* depends on the initial condition.
   (e) None of the above.

20. A fair coin is tossed 6 times. Which of the following statements is *NOT* true?
   (a) The sample space contains $2^6 = 64$ equally-likely outcomes.
   (b) The probability of obtaining exactly 4 heads is 15/64.
   (c) The probability of obtaining 0 heads is 1/64.
   (d) The probability of obtaining at most 1 head is 7/64.
   (e) The probability of obtaining exactly 3 heads is 1/2.

21. A rapid diagnostic test has sensitivity 95% and specificity 98%. The disease has prevalence 1% in the population. Given a positive test, what is the probability (rounded to two decimal places) that the patient actually has the disease?
   (a) 0.95
   (b) 0.98
   (c) 0.32
   (d) 0.05
   (e) None of the above.

22. Events A and B satisfy $P(A) = 0.4$, $P(B) = 0.3$, $P(A \cap B) = 0.1$. What is $P(A \cup B)$?
   (a) 0.7
   (b) 0.6
   (c) 0.5
   (d) 0.1
   (e) None of the above.

23. A discrete random variable X has the probability mass function shown below.

x	1	2	3	4
$P(X = x)$	0.2	0.3	0.3	0.2

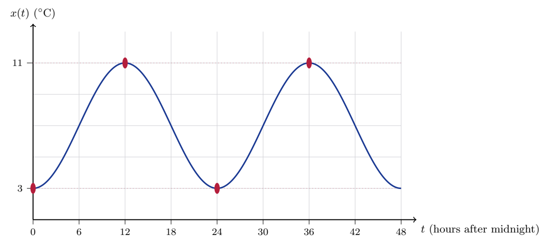
   What is the expected value $E[X]$?
   (a) 2.0
   (b) 2.5
   (c) 3.0
   (d) 1.0
   (e) None of the above.

24. Let $X \sim \text{Binomial}(n = 10,\ p = 0.5)$. What is $P(X \geq 8)$?
   (a) $\dfrac{45}{1024}$
   (b) $\dfrac{56}{1024}$
   (c) $\dfrac{11}{1024}$
   (d) $\dfrac{1}{1024}$
   (e) None of the above.

25. An existing viral variant infects 40% of contacts. Of 12 contacts of a newly-emerged variant, 9 were infected. Testing $H_0 : p \leq 0.40$ against $H_a : p > 0.40$ at significance level $\alpha = 0.05$, the p-value (from Binomial$(12, 0.40)$) is approximately 0.015. Which conclusion is correct?
   (a) Fail to reject H_0; the data are consistent with the old rate.
   (b) Reject H_0; there is statistically significant evidence that the new variant is more infectious.
   (c) Accept H_0; the old and new rates are identical.
   (d) The test is inconclusive because the sample size is too small.
   (e) None of the above.

26. Which of the following statements is *NOT* true?
   (a) 60° equals $\pi/3$ radians.
   (b) The function $y = 3\sin(4x)$ has amplitude 3.
   (c) The function $x(t) = 4\cos(2\pi t)$ has period $T = 1$ second when t is in seconds.
   (d) The function $x(t) = 4\cos(2\pi t)$ has frequency $f = 2$ Hz.
   (e) The angle $\pi/2$ radians corresponds to 90°.

27. A daily temperature oscillation is modelled by $x(t) = A\cos(B(t + C)) + D$, where t is time in hours after midnight. The temperature oscillates between 3°C and 11°C with a period of 24 hours, and $x(0) = 3$°C (i.e. the minimum occurs at midnight). The graph of $x(t)$ over two days is shown below. Which set of parameters is correct?



   (a) $A = 4,\ B = \pi/12,\ C = 0,\ D = 7$
   (b) $A = 8,\ B = \pi/12,\ C = 12,\ D = 3$
   (c) $A = 4,\ B = \pi/24,\ C = 12,\ D = 7$
   (d) $A = 4,\ B = \pi/12,\ C = 12,\ D = 7$
   (e) None of the above.

28. The sinusoidal oscillator $x(t) = 5\sin(4\pi t)$, with t in seconds, has period T and frequency f given by
   (a) $T = 0.5$ s, $f = 2$ Hz
   (b) $T = 2$ s, $f = 0.5$ Hz
   (c) $T = 4\pi$ s, $f = 1/(4\pi)$ Hz
   (d) $T = 1$ s, $f = 1$ Hz
   (e) None of the above.

29. Market demand is $Q^d = 120 - 3P$ and market supply is $Q^s = -60 + 3P$. At equilibrium, (P^*, Q^*) is
   (a) $(30, 30)$
   (b) $(20, 60)$

(c) $(40, 0)$
(d) $(30, 60)$
(e) None of the above.

30. Identify the vertical and horizontal asymptotes of the function $f(x) = \dfrac{3}{x-2} + 5$.
(a) Vertical asymptote $x = 2$; horizontal asymptote $y = 5$.
(b) Vertical asymptote $x = 5$; horizontal asymptote $y = 2$.
(c) Vertical asymptote $x = 0$; horizontal asymptote $y = 0$.
(d) Vertical asymptote $x = -2$; horizontal asymptote $y = 3$.
(e) The function has no asymptotes.

31. A new rapid antigen test has sensitivity 90% and specificity 95%. It is used to screen a student population in which the true disease prevalence is 5%. What is the *positive predictive value* (PPV)?
(a) PPV ≈ 0.90
(b) PPV ≈ 0.95
(c) PPV ≈ 0.49
(d) PPV ≈ 0.05
(e) None of the above.

32. A bacterial colony doubles every hour, starting from an initial population of 3 cells. The *total number of cells* over the first 8 hours (i.e. the sum at hours $t = 0, 1, \ldots, 7$, using $S_n = a\dfrac{r^n - 1}{r - 1}$) is
(a) 765
(b) 384
(c) 255
(d) 512
(e) None of the above.

33. In a field survey, two insect species A and B are trapped independently. The probability of capturing species A is $P(A) = 0.3$ and of species B is $P(B) = 0.4$. What is the probability of capturing *both* species?
(a) 0.70
(b) 0.12
(c) 0.10
(d) 0.00
(e) None of the above.

34. Consider the function $f(x) = x^3 - 6x^2 + 5$. Which statement about its concavity is *correct*?
(a) f has an inflection point at $(2, -11)$; concave down on $(-\infty, 2)$, concave up on $(2, \infty)$.
(b) f has an inflection point at $(2, -11)$; concave up on $(-\infty, 2)$, concave down on $(2, \infty)$.
(c) f has an inflection point at $(0, 5)$; concave up everywhere.
(d) f has no inflection point; concave up on all of $\mathbb{R}$.
(e) None of the above.

35. A loudspeaker diaphragm oscillates as $x(t) = 5\sin(4\pi t)$ (mm, t in seconds). The instantaneous velocity $x'(t)$ is
(a) $x'(t) = 20\pi\cos(4\pi t)$
(b) $x'(t) = 5\cos(4\pi t)$
(c) $x'(t) = -20\pi\sin(4\pi t)$
(d) $x'(t) = 4\pi\cos(4\pi t)$
(e) None of the above.

36. Let f be continuous and let $F(x) = \displaystyle\int_a^x f(t)\,dt$. Which of the following statements is *correct* (Fundamental Theorem of Calculus)?
(a) $F'(x) = f(x)$; differentiation and integration are inverse operations for continuous f.
(b) $F'(x) = f(a)$.
(c) $F'(x) = \int_a^x f'(t)\,dt$.
(d) $F(x) = f(x) - f(a)$.
(e) None of the above.

PART II — Python as a Mathematical Tool (14 marks; 2 marks per question)

Answer ALL questions using the MCQ answer sheet.
No computer is required. All questions are answerable from your understanding of how Python tools work.

37. Consider the following Python code:

```
from scipy.integrate import quad
import numpy as np
result, _ = quad(lambda x: x**2 * np.exp(-x), 0, 5)
```

Which of the following best describes what `result` contains after this code runs?

(a) The derivative of x^2e^{-x} evaluated at $x = 5$.

(b) The numerical value of the definite integral $\int_0^5 x^2e^{-x}\,dx$.

(c) The antiderivative $F(x) = -x^2e^{-x} - 2xe^{-x} - 2e^{-x}$ evaluated at $x = 5$.

(d) The maximum value of x^2e^{-x} on $[0, 5]$.

(e) None of the above.

38. Consider the following code:

```
import numpy as np
import matplotlib.pyplot as plt
x = np.linspace(0, 2*np.pi, 500)
y = 3 * np.sin(2*x)
plt.plot(x, y)
plt.fill_between(x, y, 0)
plt.show()
```

What does `plt.fill_between(x, y, 0)` add to the plot?

(a) A horizontal line at $y = 0$.

(b) A plot of the derivative y' as a function of x.

(c) The shaded region between the curve $y = 3\sin(2x)$ and the x-axis.

(d) A normalised area so the total shaded region equals 1.

(e) None of the above.

39. The following code numerically differentiates a function:

```
import numpy as np
x = np.linspace(0, 10, 1000)
y = 3*x**2 - 2*x + 1
dydx = np.gradient(y, x)
```

The exact derivative at $x = 4$ is $\frac{dy}{dx} = 6(4) - 2 = 22$. What does `np.gradient(y, x)` compute?

(a) The exact symbolic derivative $6x - 2$ evaluated at each point in `x`.

(b) The cumulative integral $\int(3x^2 - 2x + 1)\,dx$ at each point.

(c) A numerical approximation of dy/dx at each point in `x`.

(d) The second derivative $d^2y/dx^2 = 6$ at each point.

(e) None of the above.

40. A student writes the following code to find the steady state of a Lotka-Volterra system:

```
from scipy.optimize import fsolve
alpha, beta, lam, gamma = 0.10, 0.004, 0.0004, 0.08
def equations(vars):
    H, P = vars
    return [alpha*H - beta*H*P,
            lam*H*P - gamma*P]
H_star, P_star = fsolve(equations, [100, 20])
```

What is the code attempting to find?

(a) The maximum values of H and P over time.
(b) The equilibrium (steady-state) values (H^*, P^*) where both derivatives equal zero.
(c) The rate of change of H and P at the point (100, 20).
(d) The solution trajectory $H(t)$ and $P(t)$ starting from $(H_0, P_0) = (100, 20)$.
(e) None of the above.

41. A student is computing the p-value for a one-sided hypothesis test:

```
from scipy.stats import binom
# H0: p <= 0.40 vs Ha: p > 0.40
# n = 12 trials, 9 observed successes
p_value = 1 - binom.cdf(8, 12, 0.40)
```

Which of the following correctly describes what `binom.cdf(8, 12, 0.40)` returns?

(a) $P(X = 8)$ for $X \sim \text{Binomial}(12,\ 0.40)$.
(b) $P(X \geq 8)$ for $X \sim \text{Binomial}(12,\ 0.40)$.
(c) $P(X \leq 8)$ for $X \sim \text{Binomial}(12,\ 0.40)$, so that `p_value` $= P(X \geq 9)$.
(d) The probability density function evaluated at $x = 8$ for a Binomial distribution.
(e) None of the above.

42. Consider the following optimisation code:

```
from scipy.optimize import minimize_scalar
res = minimize_scalar(
    lambda x: x**2 - 6*x + 5,
    bounds=(0, 10),
    method='bounded'
)
```

What does this code find?

(a) The root(s) of $x^2 - 6x + 5 = 0$ on $[0, 10]$.
(b) The value of $x \in [0, 10]$ that *minimises* $f(x) = x^2 - 6x + 5$.
(c) The value of $x \in [0, 10]$ that *maximises* $f(x) = x^2 - 6x + 5$.
(d) The integral $\int_0^{10} (x^2 - 6x + 5)\, dx$.
(e) None of the above.

43. A student implements logistic growth in Python:

```
import numpy as np
K, A, alpha = 800, 99, 0.20
t = np.linspace(0, 50, 500)
P = K / (1 + A * np.exp(-alpha * t))
```

What is the value of `P[0]` (the value of `P` at $t = 0$), and what does it represent in the logistic model?

(a) $P(0) = 800$; the carrying capacity K.
(b) $P(0) = 99$; the growth parameter A.
(c) $P(0) = K/(1 + A) = 8$; the initial population size.
(d) $P(0) = 0.20$; the intrinsic growth rate α.
(e) None of the above.

PART III — Long-Answer Questions (48 marks; 12 marks per question)

Answer ALL four questions in the answer booklet. Show all working.

44. **(12 marks)** A livestock producer has 80 animals currently weighing 60 kg each. The per-animal weight-gain rate is

$$\frac{dw}{dt} = 2.5 - 0.08\,t \quad \text{(kg/day)},$$

so that weight gain slows over time. The feed cost is \$0.60 per animal per day, and the market price is \$6.00 per kilogram of live weight. The producer wishes to choose the holding period t (days) that *maximises profit* on the entire herd.

(a) (3 points) Write the weight-per-animal function $w(t)$, given $w(0) = 60$ kg.

(b) (3 points) Derive an expression for total profit $\pi(t)$ from selling the 80-animal herd after t days of feeding. (Total revenue minus total feed cost; treat the initial animal cost as sunk.)

(c) (3 points) Find the holding period t^* that maximises $\pi(t)$ and verify it is a maximum by checking the second-order condition.

(d) (3 points) Compute the optimal weight per animal, the total revenue, the total feed cost, and the maximum profit.

45. **(12 marks)** A dietitian designs a feed ration from two foods, Food I and Food II, in units x and y per animal per day. The nutrient contents, minimum daily requirements, and costs are:

Nutrient	**Food I (per unit)**	**Food II (per unit)**	**Minimum required**
Protein (g)	5	4	120
Energy (kJ)	15	20	500
Vitamin A (mg)	0.5	1	8
Cost (\$/unit)	3	2	—

Veterinary advice requires at least 4 units of Food I per day ($x \geq 4$). The goal is to *minimise daily cost* $C = 3x + 2y$ subject to all constraints (and $x, y \geq 0$).

(a) (2 points) Write out the complete linear-programming formulation (objective and all constraints).

(b) (4 points) Sketch the feasible region in the (x, y) plane, labelling all boundary lines, the feasible region, and the vertices.

(c) (3 points) Use the corner-point theorem to find the minimum-cost ration. State the optimal (x^*, y^*) and the minimum cost.

(d) (2 points) Verify the optimal ration satisfies all constraints, and identify which are *binding*.

(e) (1 point) **Sensitivity.** Suppose the cost of Food I rises from \$3 to \$6 per unit. Recompute the minimum-cost ration and new minimum cost.

46. **(12 marks)** A coastal fishery's biomass S (tonnes) obeys the Schaefer surplus-production model:

$$\frac{dS}{dt} = g\,S\left(1 - \frac{S}{K}\right) - H,$$

where $g = 0.25$ per year, $K = 800$ tonnes, and H is the annual harvest.

(a) (3 points) Assuming H is sustainable ($dS/dt = 0$), derive the *maximum sustainable yield* (MSY) and the biomass S_{MSY} at which it occurs. Show that MSY $= gK/4$ and $S_{\text{MSY}} = K/2$.

(b) (3 points) The current biomass is $S_0 = 200$ tonnes. If harvest quotas equal the MSY, compute dS/dt at $S = 200$. What happens to the stock, and what does this mean for long-term fishery viability?

(c) (3 points) What is the maximum harvest $H_{\max}(S)$ that keeps the stock at $S = 200$ stable?

(d) (3 points) If each tonne of catch sells for \$1,000, compute the annual revenue at MSY. In one sentence, explain why operating above MSY in the short term is not a viable long-term strategy.

47. **(12 marks)** A drug is given via intravenous infusion at a constant rate $k_{\text{in}} = 10$ mg/hr into a compartment of volume $V = 40$ L. The drug is cleared at rate $k_{\text{out}} = 0.2$ per hour (first-order elimination). Let $C(t)$ be the plasma concentration (mg/L) at time t (hours), with $C(0) = 0$.

(a) (3 points) **Set up the model.** Show that the concentration satisfies

$$\frac{dC}{dt} = \frac{k_{\text{in}}}{V} - k_{\text{out}}\, C = 0.25 - 0.2\, C.$$

(b) (2 points) **Steady state.** Find the steady-state concentration C_{ss} (where $dC/dt = 0$).

(c) (4 points) **Solve the ODE.** Use separation of variables to solve $dC/dt = 0.25 - 0.2\, C$ with $C(0) = 0$, and write $C(t)$ explicitly.

(d) (3 points) **Time to 90% of steady state.** Compute the time t_{90} at which $C(t_{90}) = 0.9\, C_{\text{ss}}$.

— **END OF SAMPLE EXAM** —